

Edge Data Centre — UPS for Critical Network Load

Single-Phase Uninterruptible Power Supply · Network Equipment Room

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1. Scope

This design basis governs the uninterruptible power supply serving the critical network and edge-compute load of a Tier-equivalent edge data centre / network equipment room (single-phase, ≤ 6 kW critical load). The UPS shall protect routers, switches, and edge-compute nodes through utility disturbances and short outages until the standby generator assumes load or an orderly shutdown completes.

Where a vendor proposal differs from any requirement below, the proposal shall be recorded as a deviation and is non-conforming unless approved in writing by the Engineer. Efficiency, runtime, and power-factor claims shall be substantiated by the manufacturer's published datasheet.

2. Performance Requirements

Ref	Parameter	Requirement
2.1	Topology	Online double-conversion (VFI-SS-111) . Line-interactive and standby topologies are not acceptable.
2.2	Rated capacity	≥ 6 kVA / 6 kW continuous at the stated reference conditions
2.3	Output power factor	≥ 0.9 (unity preferred)
2.4	Operating efficiency at 100% load	$\geq 96.0\%$ in online (double-conversion) mode . Efficiency achieved only in ECO/eco-mode shall NOT be accepted as evidence of compliance.
2.5	Battery autonomy at full rated load	≥ 10 minutes with internal/standard batteries, at end of design life
2.6	Input current THD	$\leq 5\%$
2.7	Availability / redundancy	N+1 (parallel-capable, so one module can be serviced without dropping the network load)
2.8	Acoustic noise at 1 m	≤ 55 dB(A)
2.9	Network management	SNMP-capable management card with remote shutdown

3. Notes to Tenderer

- A single, non-parallel unit does **not** satisfy clause 2.7.
- Quote efficiency in **online mode**, not ECO mode (clause 2.4). Many compact UPS reach their headline efficiency only in ECO/line-interactive mode, which is not the operating mode required here.
- State battery autonomy at **full rated load**; runtimes quoted at 50% load or beginning-of-life only will be treated as non-conforming evidence.

